PriorityQueue Interview Handbook

1. Why was PriorityQueue created?

A normal Queue follows FIFO, but many real-world problems require processing based on importance rather than arrival time. For example, in a hospital emergency room, a heart attack patient should be treated before someone with a mild fever. PriorityQueue solves this by removing elements according to priority.

2. Common Use Cases

CPU scheduling, hospital emergency systems, Dijkstra's shortest path algorithm, network packet scheduling, job scheduling and event processing.

3. What is PriorityQueue?

PriorityQueue removes the element with the highest priority first. By default in Java, the smallest element has the highest priority.

4. Internal Working

PriorityQueue is implemented using a Binary Min Heap stored internally in an array. It is not backed by LinkedList or TreeMap.

5. Binary Min Heap

Rule: Every parent node is smaller than or equal to its children. Therefore, the smallest element is always at the root and peek() is O(1).

6. offer() Flow

Insert the element at the end of the heap, then perform Heapify Up by comparing it with its parent and swapping until the heap property is restored. Complexity: O(log n).

7. poll() Flow

Remove the root, move the last element to the root, then perform Heapify Down by swapping with the smaller child until the heap property is restored. Complexity: O(log n).

8. peek()

Returns the root element without removing it. Complexity: O(1).

9. Time Complexity

Operation	Complexity	Reason
offer()	O(log n)	Heapify Up
poll()	O(log n)	Heapify Down
peek()	O(1)	Root element
contains()	O(n)	Linear search

10. PriorityQueue vs Queue

Queue follows FIFO. PriorityQueue removes based on priority. Queue is commonly implemented using LinkedList, whereas PriorityQueue uses a Binary Heap.

11. Custom Priority

Use a Comparator to define your own priority. For example, a reverse comparator makes the largest element the highest priority.

12. Common Interview Questions

Why is offer() O(log n)? Why is peek() O(1)? Which data structure is used internally? Why doesn't it return elements in insertion order?

13. Common Mistakes

Thinking PriorityQueue keeps all elements sorted, assuming iteration order is sorted, believing it uses LinkedList internally, and expecting contains() to be O(1).

14. Remember Forever

PriorityQueue = Binary Min Heap. Smallest element first by default. peek() O(1). offer()/poll() O(log n). Comparator changes priority.